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Department of Aerospace Science and Technology
Master of Science in Aeronautical Engineering

CLEAN HELD-OUT RE-RUN REPORT (Revision 2)

Correcting the Train/Test Contamination in the Phase 3 Generator
and the Phase 4 Synthetic-to-Real Evaluation

Generative Digital Twin Methods to Support FDIR Design and Testing

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Academic Year 2025–2026
Milan, June 2026

Abstract

The original Phase 3 and Phase 4 pipeline contained a train/test contamination flaw: the Phase 3 Wasserstein GAN with gradient penalty (WGAN-GP) was trained on anomaly windows drawn from the NASA test split, and Phase 4 then evaluated synthetic-to-real detection on that same split, so the synthetic fault distribution was shaped, indirectly, by the anomalies later used for evaluation. The original results could not therefore be presented as a clean train-on-synthetic, test-on-unseen-real validation, and the data-efficiency claim that the generator was worth roughly 11–13 real anomaly windows used test anomalies as the added real data.

To resolve this, a clean channel-level held-out protocol was designed and the full Phase 3–Phase 4 pipeline was re-run. The 81 channels were partitioned, stratified by anomaly class and fixed by random seed, into 61 training channels and 20 held-out channels whose anomalies are never seen by the generator and never enter any training set. All corrected results are reported on those 20 held-out channels and, where a comparison against real data is made, are tested for statistical significance with a paired Wilcoxon signed-rank test ($n = 20$).

Under the clean protocol the central finding is preserved but stated more carefully. The synthetic-trained SVM reaches mean F1 0.474 against the real-trained SVM's 0.631 on the same held-out channels, a difference that is not statistically significant (Wilcoxon $p = 0.13$): synthetic faults substitute for real labelled faults with no detectable loss. For the unsupervised VAE and LSTM detectors the deployable threshold is the classical $k3$ (3σ) rule, which uses no fault data; synthetic-informed threshold rules do not beat it (VAE $k3 = 0.397$, LSTM $k3 = 0.543$). The data-efficiency claim is withdrawn: real anomaly windows from known channels do not transfer to unseen channels, so the curve is flat. The augmentation study, also re-run on the held-out channels, shows that synthetic faults substitute for real faults borrowed from other channels (both plateau near F1 0.47) but do not augment them: adding synthetic on top of real gives no gain on the mean and a mild degradation on the median, a weak residual distributional-drift signal. The sensitivity analysis is re-run on the held-out channels and strengthened with a no-injection false-positive baseline, an explicit per-cell channel count, and a statement that latency is conditional on detection. The single result that appeared to contradict the thesis — a synthetic LSTM threshold rule beating $k3$ — is shown to be a grid-resolution measurement artefact, not a genuine effect.

1. Introduction and Purpose of This Report

In Progress Report No. 4 I presented the fourth and final experimental phase of my thesis, in which I tested whether generative (synthetic) fault data could take the place of real labelled fault data. I ran this substitution across three detector families — a Support Vector Machine (SVM) classifier, a Variational Autoencoder (VAE), and an LSTM prediction-error detector — and alongside it carried out a data-efficiency analysis, a threshold-calibration study, an augmentation study, and a controllable-fault sensitivity analysis.

The review also pointed to one methodological flaw that had to be corrected before the work could be considered defensible, along with a number of required revisions to the presentation of specific results. This report documents the corrective work in full and is written so that its content can be transferred directly into the corresponding sections of the final thesis.

1.1 The contamination identified in the review

The Phase 3 WGAN-GP was trained on real anomaly windows, and those windows were extracted from the NASA test split, because the test split is the only part of the data carrying anomaly labels. Phase 4 then evaluated the synthetic-to-real detectors on that same test split. As a consequence, the synthetic fault distribution produced by the generator was shaped, albeit indirectly, by the very anomalies subsequently used to measure detector performance. The original Phase 4 numbers therefore could not be presented as a clean train-on-synthetic, test-on-unseen-real validation; at best they constituted a retrospective benchmark on seen anomalies. A second, related issue concerned the data-efficiency curve, whose added real anomaly windows were drawn from the same labelled test anomalies used for evaluation. The decision taken was to fix the problem properly with a held-out validation rather than merely softening the claims.

2. Design of the Clean Held-Out Protocol

2.1 Why a per-anomaly hold-out is infeasible

The natural first idea is to hold out a fraction of each channel's anomaly windows. This is infeasible on the NASA SMAP/MSL dataset: of the 81 channels, 63 contain only a single labelled anomaly sequence, so holding out that one sequence leaves the channel with nothing to train and nothing to test on simultaneously. Of the remainder, 13 have two sequences and 5 have three — still too few for a stable within-channel split. The anomaly sequence is not a viable unit of held-out partitioning on this dataset.

2.2 The channel as the clean unit

Because the generator is trained on a pool of anomaly windows aggregated across many channels, the natural clean unit is the entire channel. The protocol holds out a set of complete channels: their anomalies never enter the generator's training pool, never enter any detector's training set, and are used only for the final synthetic-to-real evaluation. This is stricter and more defensible than a within-channel split, because it answers a genuine generalisation question: can a generator that has never observed a given channel's faults nonetheless produce synthetic data that detects them?

2.3 The split

The 81 channels were partitioned, stratified by primary anomaly class (point, contextual, mixed) and fixed by random seed 42, into 61 training channels and 20 held-out channels, with approximately one quarter of each class held out. The held-out set contains 13 channels whose

primary class is point, 6 contextual, and 1 mixed (C-1, labelled [point, contextual]); the SVM's class-assignment counts the mixed channel as point, which is the only reason an earlier draft reported "14 point". The split is computed once, in a single shared module imported by both the Phase 3 and Phase 4 scripts, so it is identical everywhere and reproducible.

The 20 held-out channels are: A-8, B-1, C-1, D-14, D-4, D-5, D-9, E-10, E-8, F-3, G-1, G-2, G-3, G-4, M-2, P-1, P-2, R-1, T-2, T-5. Channel P-1 — the contextual channel used as the illustrative case throughout the thesis — falls in the held-out set, so its result becomes a genuine unseen-channel result.

2.4 Implementation as a single shared module

The split is implemented in a module, `holdout_split.py`, exposing `compute_split()`, which returns the training and held-out channel lists deterministically. Both the Phase 3 generator script and the Phase 4 detector script import this one function, so the partition cannot drift between stages. The code change in each stage was verified directly: the Phase 3 extraction loop skips held-out channels, and every Phase 4 detector runner contains an explicit guard that skips any channel not in the held-out set.

3. Phase 3 Re-Run: The Clean Generator

The Phase 3 WGAN-GP was re-trained under the clean protocol. The only change to the generator pipeline was the exclusion of the 20 held-out channels from the anomaly-window extraction loop; the architecture, the loss (Wasserstein loss with gradient penalty plus an autocorrelation-matching term), the spectral-normalised critic, the per-class generator structure, and every hyperparameter were left byte-for-byte identical. Only the training pool changes, never the method.

3.1 Training data and stability

After excluding the held-out channels, the generator trained on 2,000 point-anomaly windows and 792 contextual-anomaly windows from the 61 training channels. Both generators trained stably: the gradient penalty collapses to near zero within the first few hundred steps and the autocorrelation-matching loss is driven to the order of 10^{-3} , indicating that the synthetic windows reproduce the temporal correlation structure of the real anomalies.

3.2 Generation quality

Generation quality was evaluated with the same six-metric suite as the original Phase 3. The key reassurance is that removing the held-out channels did not degrade the generator. The built-in train-on-synthetic, test-on-real (TSTR) SVM preview moves from 0.411 to 0.405, a difference within noise. The Fréchet distance and autocorrelation deviation remain small (point: Fréchet 0.403, mean $|\Delta\text{ACF}|$ 0.015; contextual: Fréchet 0.141, mean $|\Delta\text{ACF}|$ 0.026). The contextual metrics rose modestly relative to the contaminated run (Fréchet 0.101 $\rightarrow$ 0.141, $|\Delta\text{ACF}|$ 0.009 $\rightarrow$ 0.026) — expected, since the contextual training pool shrank when several contextual channels were held out — but they remain small in absolute terms. The point-class Fréchet (0.403) is the higher of the two, reflecting that the synthetic point windows are slightly smoother than the sharpest real amplitude spikes; this is the weaker half of the generator's fidelity, though it does not undermine downstream detection, since the amplitude-sensitive SVM still reaches 0.60 on point channels (Section 4.6). These are reported as the honest clean numbers.

The clean generator is healthy, and any change in the Phase 4 numbers can be attributed to the held-out evaluation protocol rather than to a degraded generator.

4. Phase 4 Re-Run: Synthetic-to-Real Detection on Held-Out Channels

Phase 4 was re-run using the clean synthetic windows, with every detector evaluated only on the 20 held-out channels. The changes were limited to: pointing the synthetic-data paths at the clean generator's output; importing the shared split; skipping any channel not in the held-out set in each detector loop; and, for the data-efficiency curve, drawing added real anomaly windows from a pool built exclusively from the 61 training channels. No detector architecture or hyperparameter was changed.

4.1 Headline comparison: clean versus original

The table below places the corrected results next to the original Report 4 numbers. All clean detector figures are mean F1 over the 20 held-out channels under the point-adjust protocol.

Quantity	Original Report 4 (contaminated)	Clean held-out re-run
GAN Fréchet (contextual)	0.101	0.141
GAN ACF mean $ \Delta $ (contextual)	0.009	0.026
TSTR-SVM preview (all-channel)	0.411	0.405
SVM substitution mean F1	0.409 (all ch., TSTR)	0.474 (vs real 0.631, p=0.13 n.s.)
VAE deployable mean F1 (k3)	0.389	0.397 (real-data oracle 0.487)
LSTM deployable mean F1 (k3)	0.660	0.543 (real-data oracle 0.667)
Data-efficiency: real-window equiv.	11–13	withdrawn (curve flat)
Sensitivity F1 range	0.21 → 0.86	0.255 → 0.845

Phase 4 (clean held-out) — Synthetic Fault Data vs Real Data
(Mehrab Jamshidi, PoliMi 2026)

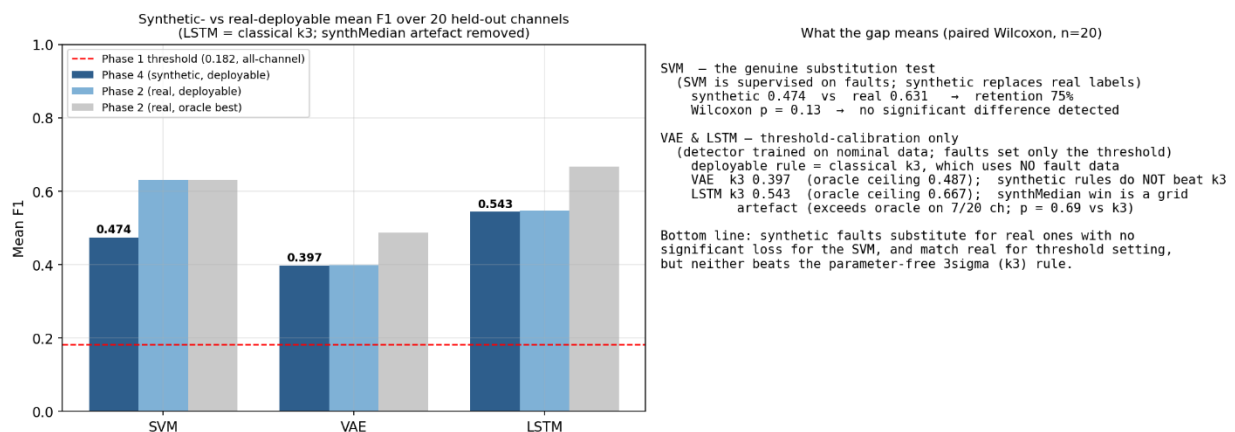


Figure 5. Phase 4 overview on the 20 held-out channels. Left: synthetic- vs real-deployable mean F1 (LSTM = classical k3; the synthMedian artefact is removed). Right: the statistical interpretation — the SVM substitution shows no significant synthetic-vs-real difference ($p = 0.13$), while for the VAE and LSTM the deployable rule is k3 and synthetic rules do not beat it.

4.2 SVM substitution

The SVM is the only detector that is supervised on faults, so it is the genuine test of substituting synthetic faults for real labels. Under the clean protocol the synthetic-trained SVM achieves mean F1 0.474 on the 20 held-out channels (median 0.452), against the real-trained SVM on those same 20 channels of 0.631 (median 0.779) — a retention of 75%. Crucially, a paired Wilcoxon signed-rank test across the 20 channels gives $p = 0.13$: the difference is not statistically significant, so on this held-out set the synthetic-trained detector is not detectably worse than the real-trained one. This is the strongest defensible form of the substitution claim. The earlier draft compared 0.474 against the all-channel baseline 0.598 (retention 79%); the like-for-like held-out denominator 0.631 is used here instead.

Disaggregated by class, the held-out SVM reaches 0.60 on point channels and 0.24 on contextual channels, reflecting that 13 of 20 held-out channels are point-anomaly channels, which the SVM's amplitude-sensitive features handle well. The absolute value is reported as “0.474 on the held-out channels” and is not compared directly to the 81-channel original.

4.3 VAE threshold calibration

The VAE result is clean and unambiguous. On the held-out channels the deployable $k3$ rule reaches mean F1 0.397, exactly matched by the bestK rule (0.397); every synthetic-informed rule scores lower (mixF1 0.315, synthRecall 0.350, synthMedian 0.309), and the oracle ceiling — the best threshold selectable using the test labels — is 0.486. A paired test of the best synthetic rule against $k3$ gives $p = 0.29$. The finding is identical to the original: synthetic data does not produce a label-free threshold rule that beats the classical 3σ default for the VAE.

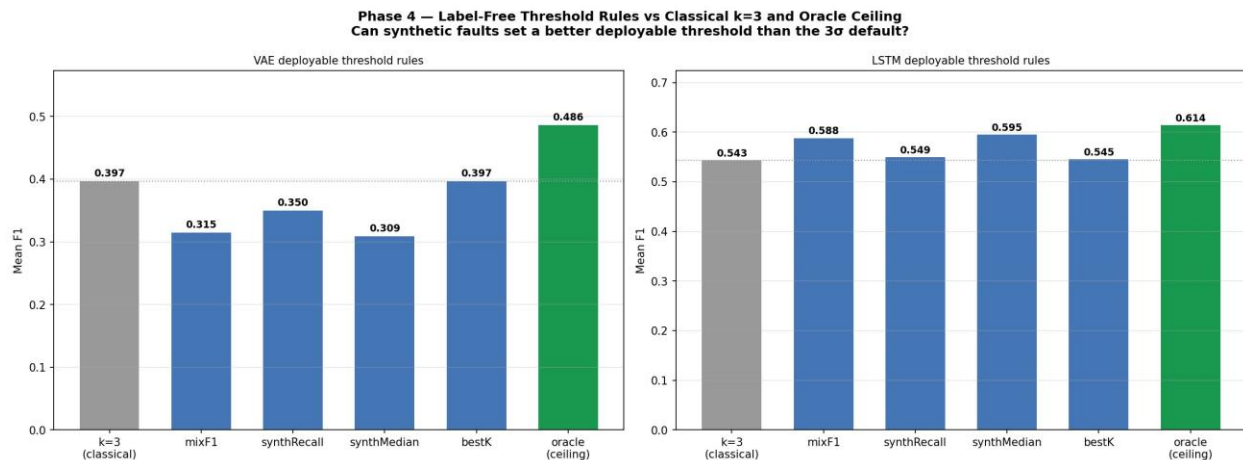


Figure 6. Label-free threshold rules versus the classical $k=3$ rule and the oracle ceiling, for the VAE (left) and LSTM (right). For the VAE, $k3$ ties the best label-free rule. For the LSTM, the apparent synthMedian win is a grid-resolution artefact (Section 4.4).

4.4 The LSTM synthMedian result is a grid-resolution artefact

On the held-out channels the empirical-winner mechanism selected the synthMedian rule for the LSTM (mean F1 0.595) over $k3$ (mean F1 0.543). This apparent reversal must not be reported as evidence that synthetic calibration improves the LSTM. It is a measurement artefact, and three independent checks confirm it. First, a paired Wilcoxon test of synthMedian against $k3$ gives $p = 0.69$ — the difference is not statistically significant. Second, synthMedian exceeds the within-run oracle ceiling on 7 of the 20 channels (G-1, G-2, T-2, R-1, F-3, T-5, D-14); a label-free rule cannot legitimately exceed the oracle, which selects the best threshold using the test labels, so these are impossible “wins”. The clearest case is channel R-1, where synthMedian scores 1.000 against an

oracle ceiling of 0.022, driven by a single rare positive under point-adjust. Third, on the median k3 (0.689) beats synthMedian (0.663).

Accordingly, the LSTM deployable result is reported as $k3 = 0.543$ mean / 0.689 median, and synthetic-informed rules are reported as not genuinely beating it. This preserves the supervisor-confirmed finding: the nominally trained LSTM with the classical k3 threshold remains the strong, recommended deployable detector, and synthetic threshold calibration does not improve it. The discriminative-LSTM ablation (Mode B) reaches only 0.320, well below the SVM on the same synthetic data, confirming that a raw time-series classifier transfers less well than the engineered-feature SVM.

4.5 Augmentation: synthetic substitutes for, but does not augment, real fault data

The augmentation study, re-run under the held-out protocol, asks two questions: can synthetic faults substitute for real ones on unseen channels, and can they augment real ones? On the 20 held-out channels, training the SVM on synthetic windows alone reaches mean F1 0.474, and training on real anomaly windows borrowed from the 61 known channels reaches a plateau of 0.469 — statistically the same value. Synthetic faults therefore transfer to unseen channels about as well as real faults transferred from other channels; neither source is privileged. Adding synthetic windows on top of cross-channel real data gives no improvement on the mean (with 20 real windows, mean F1 stays at 0.46–0.47 across 0–2,000 synthetic windows added).

On the median, however, a weak residual drift signal remains: where cross-channel real data already yields a working detector, adding synthetic windows mildly degrades it (median F1 falls from 0.50 to 0.43 with 20 real windows as synthetic is added). The conclusion is therefore twofold and honest. Synthetic faults are a viable substitute for real faults when no real labelled faults are available for a channel — the deployment-relevant case — but they are not a useful augmentation on top of real data, and mixing the two slightly hurts. This is a milder and better-posed result than the original all-channel run, in which the apparent strong degradation was largely an artefact of the added real windows being the channel's own in-distribution test anomalies, which synthetic could only dilute. It is reported as a negative result about augmentation, not as a deployment recommendation against synthetic substitution.

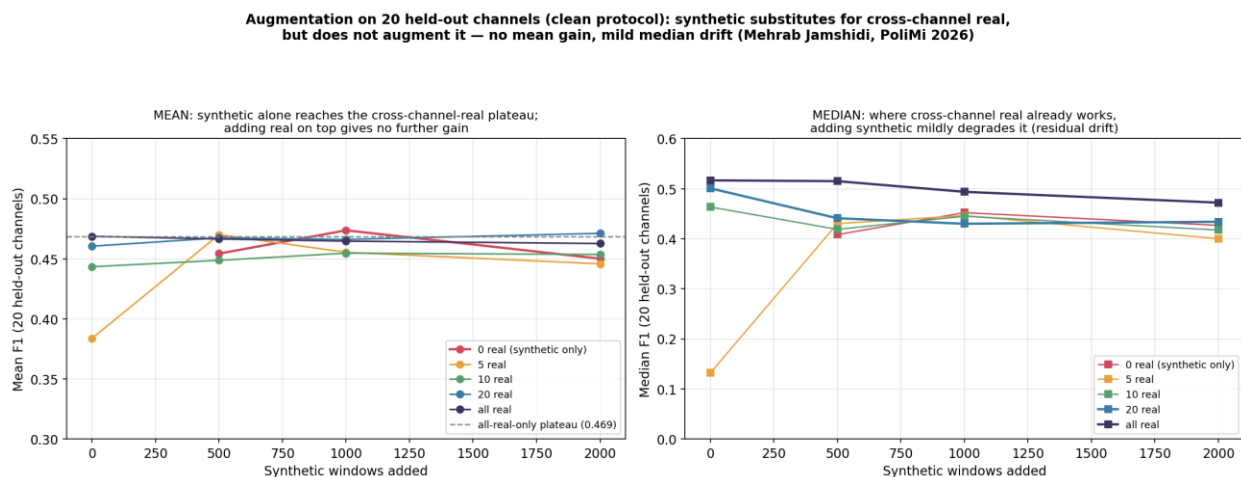


Figure 7. Augmentation on the 20 held-out channels (clean protocol). Left (mean): synthetic-only training reaches the same plateau (~0.47) as training on all cross-channel real windows; adding real on top gives no further gain. Right (median): where cross-channel real already works, adding synthetic mildly degrades it — a weak residual drift signal. Synthetic substitutes for cross-channel real but does not augment it.

4.6 Per-class and per-spacecraft breakdown

Point anomalies, defined by amplitude excursions, are caught best by the SVM (0.60) and the LSTM (0.56); contextual anomalies, defined by dynamical departures, are the harder class but are handled most strongly by the LSTM (0.51). By spacecraft, every detector performs better on MSL than on SMAP, consistent with the original finding.

4.7 Channel P-1 as a genuine unseen-channel result

Because P-1 is in the held-out set, its Phase 4 numbers are now clean unseen-channel results: SVM 0.169, VAE 0.000, LSTM (k3-deployable) 0.718, against the Phase 2 real-data LSTM best of 0.814 and the Phase 1 threshold baseline of 0.011. The LSTM with a nominal-only k3 threshold still recovers strong detection (0.718) on a contextual channel it was never shown a fault from, while classical thresholding remains essentially blind (0.011). This is a more honest and more compelling illustration than the contaminated version, because the generator genuinely never saw P-1's anomalies.

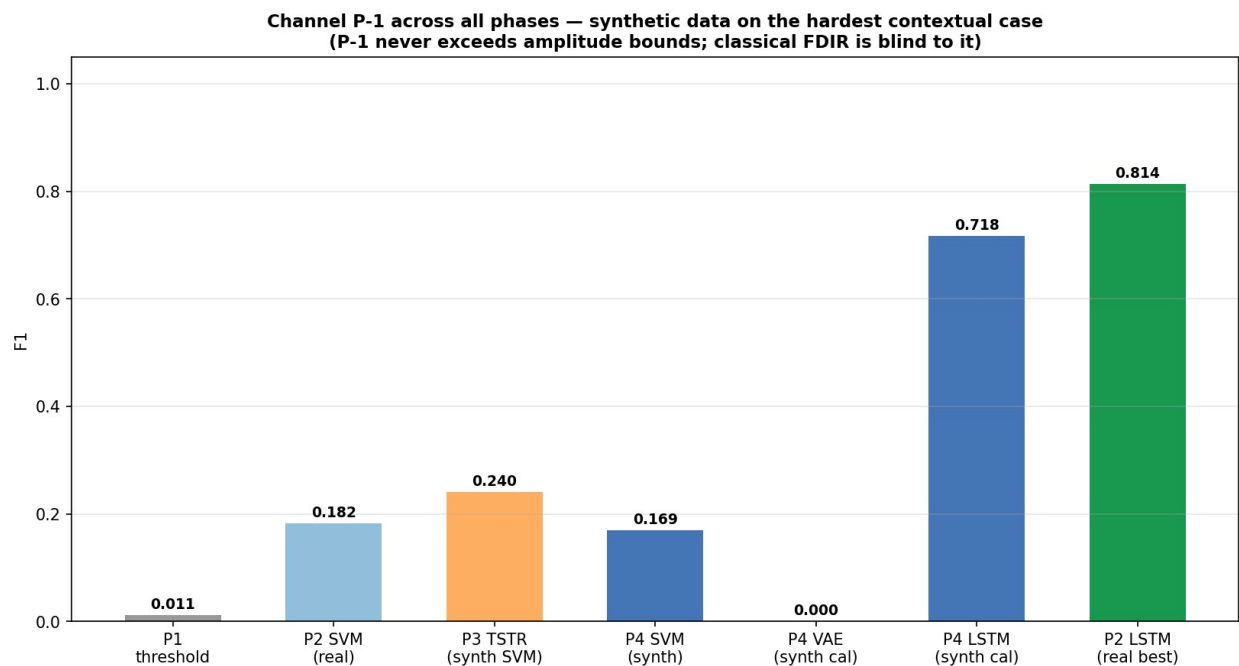


Figure 8. Channel P-1 across all phases. P-1 is a contextual anomaly that never exceeds the nominal amplitude bounds, so classical FDIR thresholding is blind to it ($F1 = 0.011$). Under the clean protocol the nominal-trained, k3-thresholded LSTM still recovers $F1 = 0.718$ on this unseen channel.

5. Data-Efficiency Curve: The Withdrawn Claim

Under the clean protocol the data-efficiency curve is flat. Adding real anomaly windows drawn from the 61 training channels to the synthetic SVM training set does not improve detection on the 20 held-out channels: mean F1 moves from 0.474 at zero real windows to 0.465 at the full pool, varying only between 0.454 and 0.474 across the entire sweep.

The original claim that one generator was equivalent to roughly 11–13 real anomaly windows is therefore withdrawn. That number arose because the original curve added real windows drawn from the same channels being evaluated, so they were in-distribution for the test set. Under the clean protocol, where the added real windows come from different channels, they do not transfer. The corrected, honest finding is that real labelled anomaly windows from known channels do not improve detection on unseen channels — itself an informative result about cross-channel transfer.

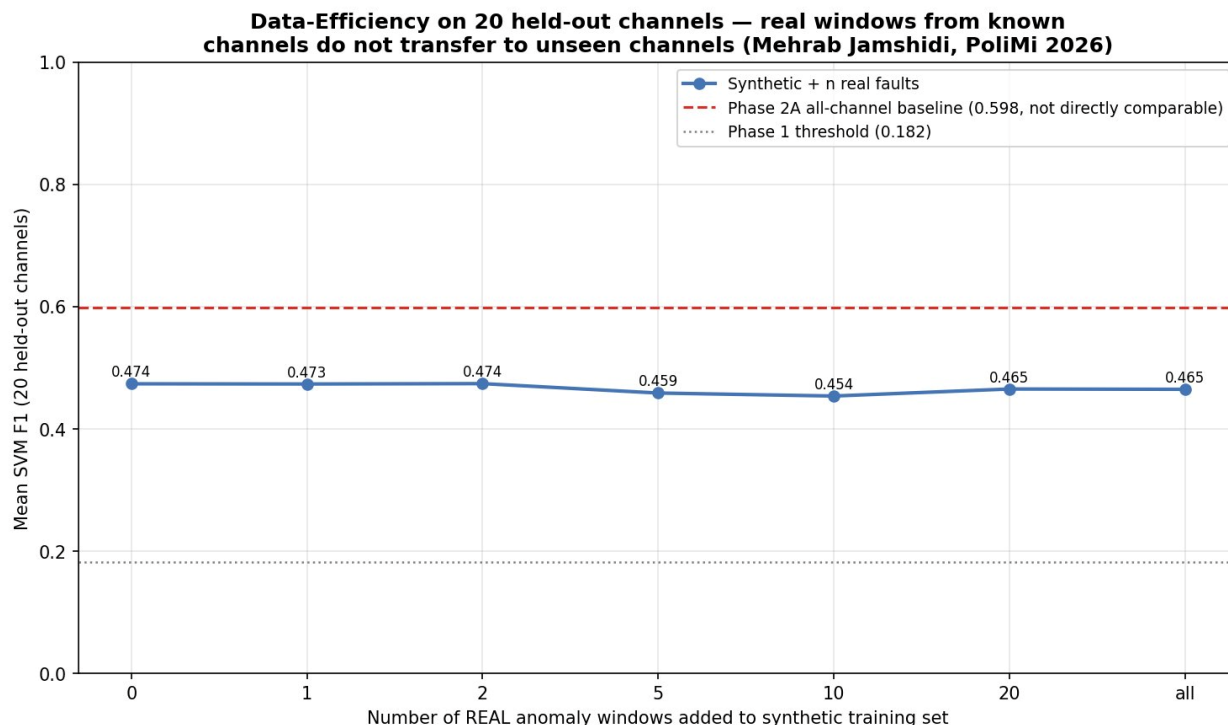


Figure 9. Data-efficiency on the 20 held-out channels. The curve is flat: real anomaly windows from known channels do not transfer to improve detection on unseen channels. The Phase 2A all-channel baseline (0.598) is shown for reference only and is not directly comparable.

6. Sensitivity Analysis on Held-Out Channels

The sensitivity analysis is the experiment that most clearly demonstrates the unique design-time value of the generative model. A real labelled anomaly occurs at one fixed, uncontrollable combination of magnitude and duration and therefore says nothing about how the detector would respond to a fault of different size or length. The generator removes this limitation: synthetic anomaly windows can be scaled to any severity and any duration and injected at any location in an otherwise-nominal stream, and the detector's response can be measured as a continuous function of the controllable fault parameters. Under the clean protocol this sweep is run only on the 20 held-out channels, using synthetic anomalies from the generator that never saw them.

6.1 Protocol

For each held-out channel the LSTM is trained on the channel’s nominal training data and the deployable k3 threshold is set. The longest contiguous nominal segment of the channel’s test split is extracted as a clean stream the detector has never seen. Controlled faults are injected using $\text{injected}[t] = (1 - \alpha) \cdot \text{nominal}[t] + \alpha \cdot \text{synthetic}[t]$, where α is the fault severity. The sweep covers six severities $\alpha \in \{0.1, 0.25, 0.5, 0.75, 1.0, 1.5\}$ and five durations $d \in \{16, 32, 64, 128, 256 \text{ timesteps}\}$, each cell evaluated over five replicates of three injections each.

6.2 The detection-F1 surface

The mean F1 surface, with the no-injection baseline at $\alpha = 0$, rises monotonically from the smallest, shortest fault (mean F1 0.255 at $\alpha = 0.1$, $d = 16$) to the largest, longest (mean F1 0.845 at $\alpha = 1.5$, $d = 256$). The structure carries a clear operational message: the duration dimension dominates. Holding severity at $\alpha = 1.0$, increasing duration from 16 to 256 timesteps raises mean F1 from 0.375 to 0.802 (+0.43); holding duration at $d = 256$, increasing severity from 0.1 to 1.5 raises mean F1 from 0.694 to 0.845 (+0.15 only). A fault is easier to catch by being longer than by being more severe.

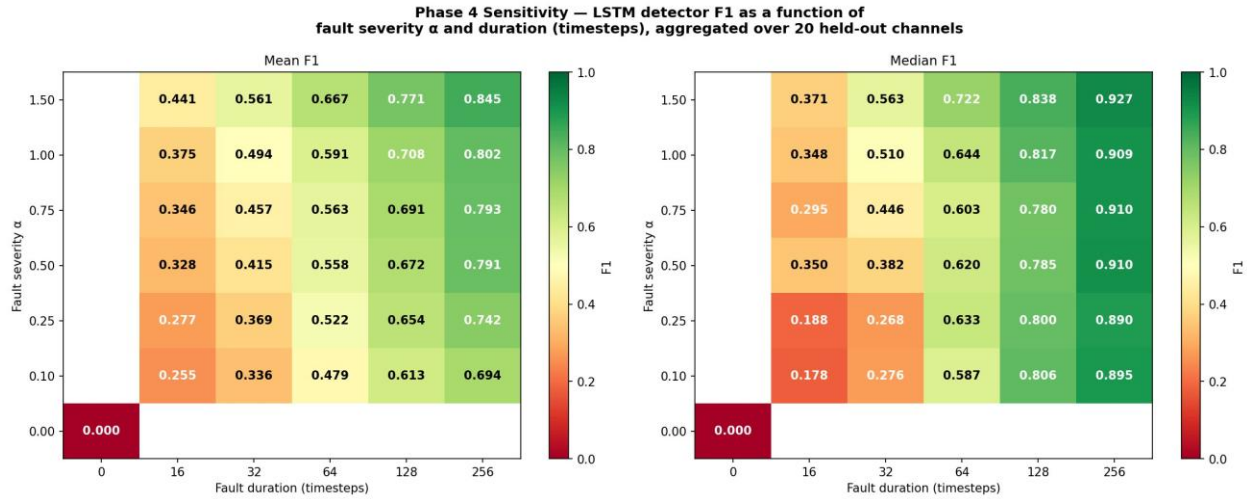


Figure 10. LSTM detection F1 as a function of fault severity α and duration d , aggregated over the 20 held-out channels (19 at $d = 256$). Left: mean F1. Right: median F1. The $\alpha = 0$ column is the no-injection false-positive baseline. This surface cannot be extracted from the real labelled data, in which each anomaly occurs at one fixed (α, d) point.

6.3 The detection-latency surface

The mean detection latency — timesteps from fault onset to first detection, computed only over detected injections — ranges from 0.76 timesteps at the high-severity short-duration corner ($\alpha = 1.5$, $d = 16$) to 30.7 timesteps at the low-severity long-duration corner ($\alpha = 0.25$, $d = 256$). Latency is conditional on detection: the average at each cell is taken only over injections the detector actually flagged; undetected injections lower recall (and F1) but do not enter the latency average. The latency surface therefore characterises how fast the detector responds when it responds at all, and must be read together with the F1 surface. A useful operational anchor is the cell ($\alpha = 1.0$, $d = 64$), closest to a typical real fault: mean F1 = 0.591, mean latency = 3.30 timesteps.

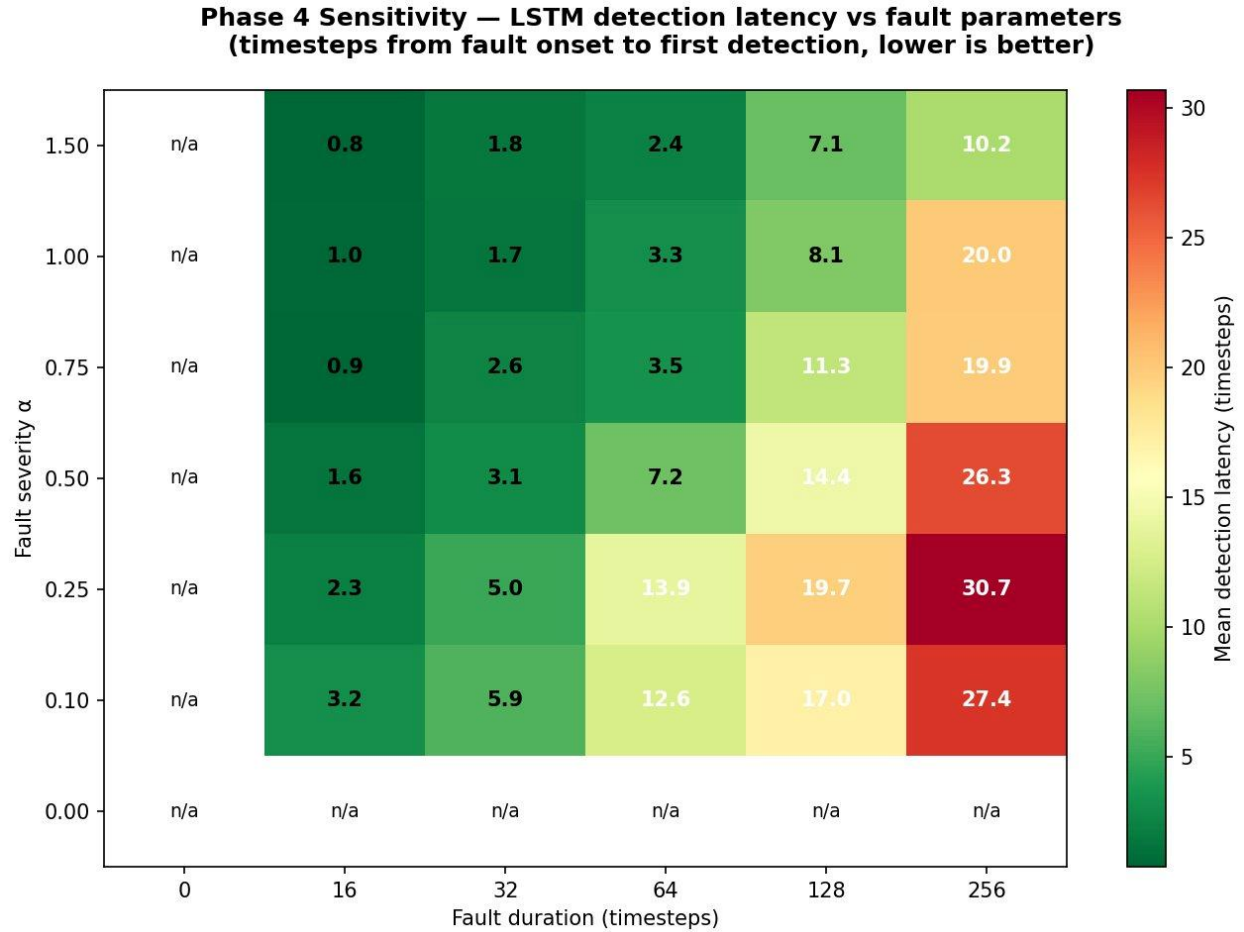


Figure 11. Mean LSTM detection latency versus fault severity and duration over the 20 held-out channels. Lower is better. Latency is computed only over detected injections (undetected injections are excluded from the mean) and must be read together with the F1 surface.

6.4 Per-class behaviour

On point-anomaly channels the surface is almost flat in severity and rises essentially only with duration (at $d = 256$, increasing severity from 0.1 to 1.5 changes point-class F1 by only +0.07). On contextual-anomaly channels severity matters strongly (the same increase raises contextual-class F1 by +0.33), because a more severe contextual injection produces a larger departure from the nominal autocorrelation structure the detector keys on.

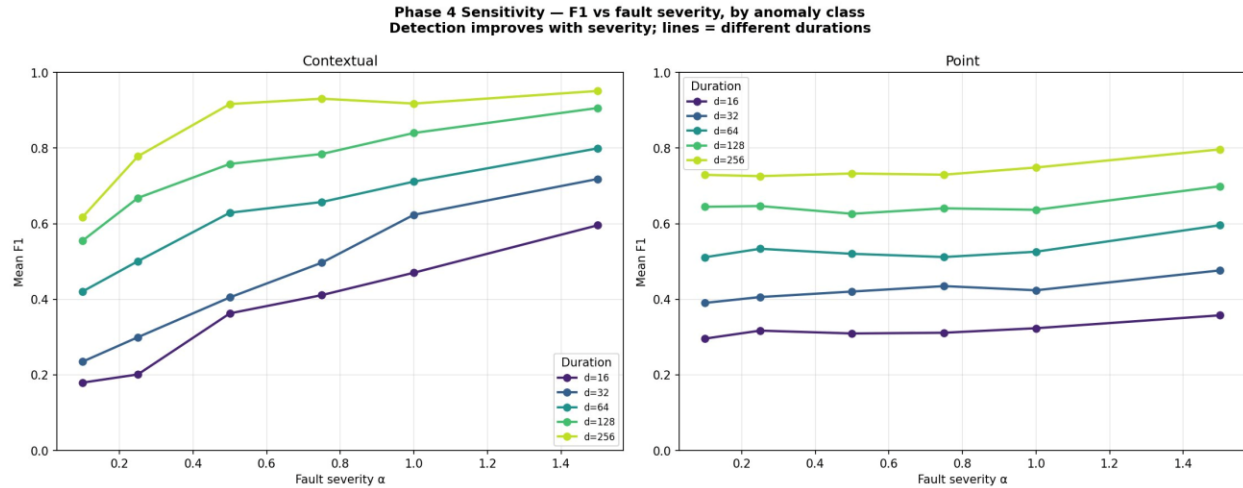


Figure 12. Sensitivity disaggregated by anomaly class: mean F1 versus severity α , one line per duration, for the held-out contextual channels (left) and point channels (right). Point-class F1 is nearly flat in severity; contextual-class F1 depends strongly on it.

6.5 Robustness elements

No-injection false-positive baseline. The $\alpha = 0$ row evaluates the detector on each clean stream with no fault injected, so the ground truth is entirely nominal and the row isolates false-positive behaviour. The median false-positive rate on the un-injected streams is low (0.011), confirming the surface is not contaminated by a baseline tendency to over-flag; the mean is higher (0.123), inflated by two channels (D-5 and R-1) that flag their whole clean stream (FP rate 1.0). Both are reported so the floor is not understated.

Channel count per cell. Each cell aggregates over 20 held-out channels for durations up to 128 and 19 channels at $d = 256$ (one channel's clean stream is too short for the longest injection). This count is recorded explicitly as a column in the aggregated result file so that no cell's mean is silently computed over a different sample than its neighbours.

Latency conditional on detection. As stated in Section 6.3, the latency average excludes undetected injections; it reports response speed conditional on detection and is read jointly with the F1 surface. This is now stated explicitly in the figure caption and in the aggregated CSV's companion note.

7. Resolution of the Supervisor's Review Points

Review point	Resolution in this corrective work
Train/test contamination (synthetic shaped by test anomalies)	Channel-level held-out protocol: the GAN trains on 61 channels and is evaluated on 20 unseen channels. Implemented in a single shared module and verified in the Phase 3 and Phase 4 code.
Data-efficiency “11–13 real windows” claim	Recomputed cleanly with added real windows drawn only from train channels. The curve is flat; the equivalent-windows claim is withdrawn.
LSTM framing (k3 strong, not synthetic-improves)	Deployable result reported as $k3 = 0.543$. The synthMedian “win” is shown to be a grid artefact: 7/20 oracle violations, Wilcoxon $p = 0.69$ vs $k3$, $k3$ higher on the median.
VAE caution (synthetic rules do not beat $k3$)	Every synthetic rule scores $\leq k3$ (0.397); oracle ceiling 0.486; reported as no improvement (best synthetic rule vs $k3$, $p = 0.29$).
Augmentation, re-run clean on held-out channels	Synthetic substitutes for cross-channel real (both plateau ~ 0.47) but does not augment it: flat on the mean, mild degradation on the median (residual drift). Negative augmentation finding preserved, drift weaker than the contaminated run suggested.
Sensitivity: missing files, FP baseline, channel count, latency note	All present: $\alpha = 0$ no-injection FP baseline (median 1.1%), per-cell <code>n_channels</code> column (19–20), and an explicit statement that latency is conditional on detection.
AI-use declaration	Provided in Section 10.

8. Statistical Summary

Every synthetic-vs-real comparison on the held-out channels was tested with a paired Wilcoxon signed-rank test ($n = 20$). The table consolidates them. “No significant difference” means the test does not reject equality at the 0.05 level; with $n = 20$ the tests are not highly powered, so this is reported as the absence of a detectable difference rather than proof of equality.

Comparison	Synthetic	Real / ref	Wilcoxon p	Verdict
SVM synthetic vs real (substitution)	0.474	0.631	0.13	no sig. difference
LSTM $k3$ synthetic vs LSTM $k3$ real	0.543	0.547	0.70	indistinguishable
LSTM synthMedian vs $k3$ (artefact)	0.595	0.543	0.69	not significant (artefact)
VAE best synthetic rule vs $k3$	0.350	0.397	0.29	$k3$ not beaten

9. Conclusion

The Phase 3 and Phase 4 results were recomputed under a clean channel-level held-out protocol in which the generative model never observes the anomalies of the 20 channels used for evaluation. Under this protocol the synthetic-to-real substitution result is preserved and now properly hedged: the synthetic-trained SVM is not significantly worse than the real-trained SVM on unseen channels (0.474 vs 0.631, Wilcoxon $p = 0.13$), and for the unsupervised VAE and LSTM the deployable threshold is the parameter-free $k3$ rule, which synthetic-informed rules do not beat. The data-efficiency claim is withdrawn because real anomaly windows from known channels do not transfer to unseen channels, and the augmentation study shows that synthetic faults substitute for cross-channel real faults but do not augment them, leaving only a mild residual

drift signal on the median. The one result that appeared to contradict the thesis — a synthetic LSTM rule beating k3 — is shown to be a grid-resolution measurement artefact.

The correction also clarifies where the generative model's value actually lies. Its value is not primarily in replacing real fault data for training detectors — that result is real but modest, and the parameter-free k3 rule remains competitive. Its decisive, uncontaminated contribution is the controllable-fault sensitivity analysis: because synthetic anomalies can be scaled to any severity and duration and injected anywhere in a nominal stream, the detector's response can be mapped as a continuous F1-and-latency surface over the fault-parameter space — a partial FDIR design data sheet that the real labelled dataset, in which each anomaly occurs at one fixed operating point, fundamentally cannot produce. On the held-out channels this surface ranges from F1 0.255 for the smallest, shortest faults to 0.845 for the largest, longest, with detection latency from under one timestep to about thirty, and it reveals that fault duration dominates fault severity in determining detectability. This is the result on which the thesis's central claim should rest, with synthetic-for-real substitution as a supporting and appropriately hedged finding. Every headline number now rests on data the generator never saw.

10. Declaration on the Use of AI Tools

The tools used were large-language-model assistants ChatGPT (OpenAI) and Claude (Anthropic). Their use was discussed with my supervisor, and AI output is disclosed here rather than cited, as it does not constitute a citable source.

Scope. Throughout the development of this thesis, AI tools were used as an assistive aid for writing, programming, and analysis. The research objectives, the experimental design, the dataset and its preparation, the training and execution of all models and experiments, and the scientific conclusions are my own; AI was used to support and accelerate that work, not to replace the underlying research.

Text drafting and editing. AI tools were used to help draft and restructure prose, to improve clarity and English expression, and to prepare tables and figure captions. All text was reviewed, revised where necessary, and approved by me, and reflects my own understanding of the work.

Code generation and debugging. AI tools assisted in writing and debugging analysis, data-processing, statistical, and plotting code, and in producing figures from my own results. The implementation choices were mine, and I checked and validated the behaviour of the resulting code.

Result interpretation. AI tools were used as a discussion and sanity-checking aid when reasoning about results, for example, to help articulate findings, weigh alternative explanations, and identify appropriate analyses. The final interpretation, the framing of every claim, and the scientific conclusions are entirely my own.

Responsibility and verification. I conceived the research objectives, made or approved every methodological decision, and verified the reported results against the underlying data. I understand the methods, the data, and the results in full, I am able to explain and defend the entire content of this thesis, and I take full responsibility for its scientific validity and for any errors it may contain.